EDSGN100 Design Project #2
Final Design Report

Underground Rare Earth Element Mine

Introduction to Engineering Design
EDSGN 100 Section 21

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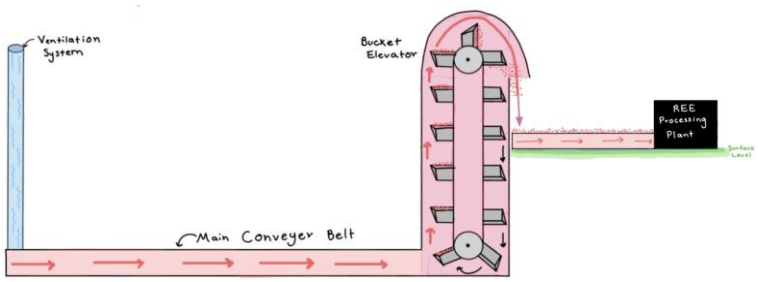

Submitted to: Prof: Colledge

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Executive Summary

General Electric provided our team with the task to create a mining system in order to extract Rare Earth Elements from ore deposits found in the Pocono Mountains. The US is in need of this ore deposit because it provides an important source of REE in many ways. The project objectives were provided by GE's extraction strategy plan, which includes using a block-caving method for the mining system. We created a list of specifications that is geared towards using cost-efficient, environmentally friendly, and safe methods of mining. After generating several designs, we chose our final design that includes a conveyor belt system that leads to a bucket elevator, which takes the ore to the surface. We performed thorough calculations to determine the total revenue obtain a net profit of $5,861,951,758,931.51.

Underground Rare Earth Element Mine

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1.0 Introduction

We were tasked with creating a process to mine Rare Earth Elements from the Pocono Mountains. General Electric (GE) sponsored this fictitious mining project and provided the details of the project with the corresponding specifications. Though Rare Earth Elements do not occur in the Pocono Mountains, GE charged each Design Team (DT) with researching and developing a process to extract the ore from the mine. GE gave specifications for the preexisting mine that the various DTs had to then build off of. GE stated that the mine had 100 draw points and a 15.3 km Haul Truck Ramp and gave information for the various costs and approximate volumes/densities of the ore.

Throughout the project, the EDSGN JEMs stayed consistently dedicated to creating the best possible product, keeping in mind the provided criteria. We began by weighing the different options for the various mining processes, eventually deciding to follow GE's suggestion of Block Caving Mining. Our group then weighed how we wanted to transport the ore from the draw points, and using a pairwise chart and a weighted matrix, we were able to determine which methods to pursue.

1.1 Design Principles

We made sure to comply with the eight specifications provided by GE, Penn State and ABET, but we emphasized several of the specifications that GE gave us. Our group decided that Airborne Contamination, Occupational Health and Workplace Safety, and Operating Costs and Productivity were the most vital, and our pairwise chart backed this up.

1.2 Gantt Chart

		1.0 Begin Project					
✓			1.1 Listen to Lecture/attend meeting		1 day	Mon 3/13/17	Mon 3/13/17
✓			1.2 Understand GE rubric requirements		1 day	Tue 3/14/17	Tue 3/14/17
✓			1.3 Modify Team Charter		1 day	Wed 3/15/17	Wed 3/15/17
✓			1.4 Create Problem Statement		1 day	Wed 3/15/17	Wed 3/15/17
		2.0 Define Problem					
✓			2.1 Customer Needs		1 day	Fri 3/17/17	Fri 3/17/17
✓			2.2 Research Requirements and Restraints		2 days	Fri 3/17/17	Mon 3/20/17
✓			2.3 Research aspects of mining		4 days	Wed 3/22/17	Mon 3/27/17
✓			2.4 Revise Problem Statement		1 day	Wed 3/22/17	Wed 3/22/17
		3.0 Generate Concepts					
✓			3.1 Idenify constraints and specs		2 days	Mon 3/20/17	Tue 3/21/17
✓			3.2 Research similar products		2 days	Fri 3/24/17	Mon 3/27/17
✓			3.3 Brainstorm Ideas		3 days	Thu 3/23/17	Mon 3/27/17
✓			3.4 Generate Concepts		3 days	Mon 3/27/17	Wed 3/29/17
✓			3.5 Analyze Concepts		1 day	Wed 3/29/17	Wed 3/29/17
✓			3.6 Select promising concepts		1 day	Fri 3/31/17	Fri 3/31/17
		4.0 Begin Detailed Design					
			4.1 Perform detailed analyses of concepts		2 days	Fri 3/31/17	Mon 4/3/17
			4.2 Sketch ideas of product		3 days	Fri 3/31/17	Tue 4/4/17
			4.3 Identify Final Method and Parts		2 days	Wed 4/5/17	Thu 4/6/17
			4.4 Determine materials that are needed and costs		3 days	Wed 4/5/17	Fri 4/7/17
			4.5 Develop detailed mining process		5 days	Fri 4/7/17	Thu 4/13/17
			4.6 CAD Drawings/Part Drawings		5 days	Fri 4/7/17	Thu 4/13/17
			4.7 Generate Final Design Drawings		5 days	Fri 4/7/17	Thu 4/13/17
		5.0 Project Report					
			5.1 Combine/Confirm Project Components		1 day	Thu 4/13/17	Thu 4/13/17
			5.2 Generate Project Report		4 days	Mon 4/17/17	Thu 4/20/17
			5.3 Generate Powerpoint Presentation		2 days	Fri 4/14/17	Mon 4/17/17

Figure 1 Gantt Chart

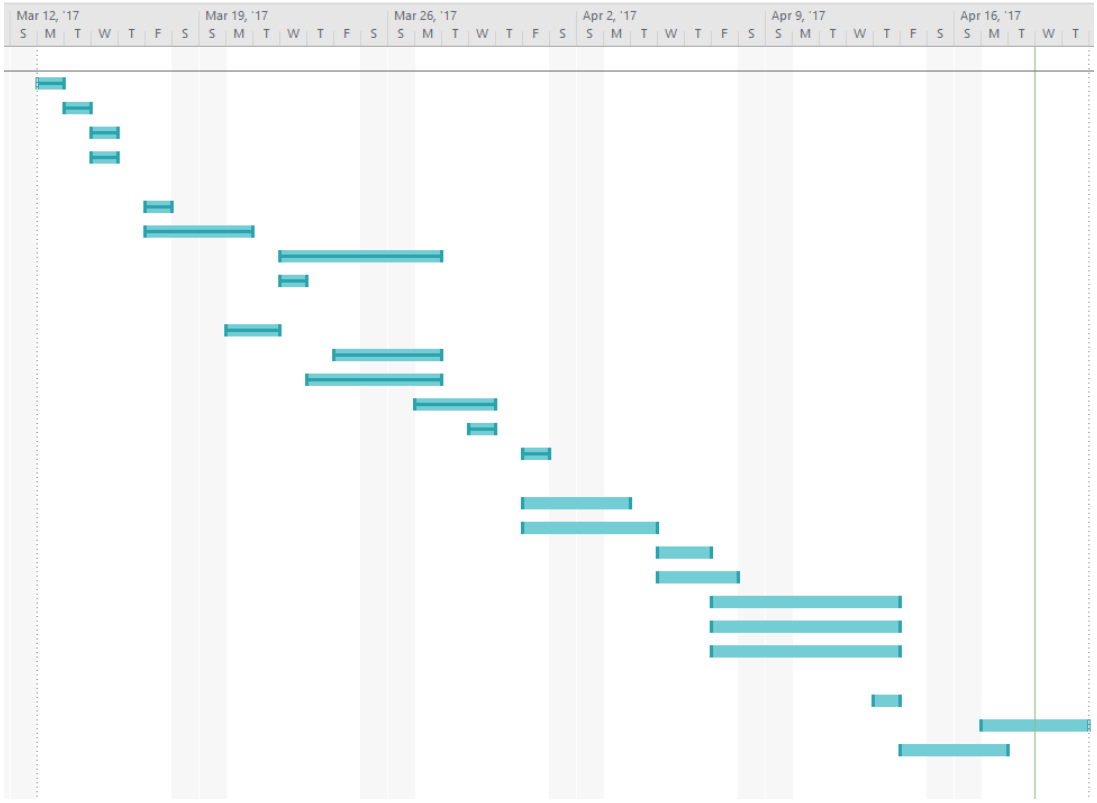

Figure 2 Gantt Chart Cont.

2.0 Project Background

The ore deposit is located in the Pocono Mountains and contains a concentration of approximately 10% Rare Earth Element per ton of ore. The deposit is sitting beneath a valley, and the top of the ore is 3000 meters underground. GE has a preexisting (fictitious) mine, and charged the DT with the task of developing a process to remove ore in a cost-efficient, environmentally friendly manner. GE approached students to help improve the Pocono mine's current techniques. The Penn State College of Engineering students, using the engineering design skills they have been taught, created a process to mine the ore in a safe, environmentally conscious, cost efficient manner. Students needed to take into account measures to improve these factors, including using different types of mining equipment or adding onto the existing mine.

Using data gathered from the GE: Mining Statement of Work as well as the FAQ page on the EDSGN website, Penn State students were able to determine the constraints and develop a feasible solution. Students were at liberty to decide what type of mining, type of transport from draw points, type of transport to the surface, and other stipulations that were left open for interpretation.

We were given information on the terrain of the Pocono Mountains (rock types include sandstone, siltstone and conglomerate); the current setup of the mine (Block Caving Mining with 100 draw points, each 30 meters away from a Haul Truck Loading point); and the type of equipment they were planning on using (Load Haul Dumps and Haul Trucks, most likely diesel). With the specifications provided by GE, we were able to design the process for mining Rare Earth Elements in order to produce the desired outcome.

3.0 Project Objectives

Our project objectives were provided from GE's extraction strategy plan and from a presentation given by GE representatives. We were tasked to design a mining system in order to extract Rare Earth Elements from ore deposits found in the Pocono Mountain range. The ore would be a great source of REE for industrial applications and a major source of income for mining companies. Before developing our strategy, we had to satisfy the eight stakeholder needs provided by GE. They include airborne contamination, stewardship, environmental management, occupational health and workplace safety, operating costs and productivity, local community engagement, waste management and disposal, and mine closure and rehabilitation. We were required to use the block-caving method because it has a lower cost per ton compared to other mining systems. Considering both the stakeholder needs and restraints, we had to design a mining system to extract the REE from the ore deposits.

3.1 Problem Statement

Our goal is to create a mining process that utilizes cost-efficient methods, taking into account environmental factors, health and safety concerns, and ore processing in order to extract Rare Earth Elements from deposits found 3,000 meters under the earth.

3.2 Specifications

Stakeholder Need	Specification
1. Airborne Contamination	• The percentage of airborne contaminations in the mine's air cannot exceed levels of: ○ Diesel fumes - 160 µg/m^3 carbon ○ Carbon Monoxide (CO) - 2.5% ○ Hydrogen (H) - 0.8% ○ Hydrogen Sulfide (H2S) - .8% ○ Acetylene (C2H2) - 0.4% ○ Propane (C3H8) - 0.4% ○ Sulfur Dioxide (S02) - 5.0 ppm ○ Methane (CH4) - 1.0%
2. Stewardship	• 90% of REE is extracted from ore deposits
3. Environmental Management	• Mine waste and runoff will be minimized to meet appropriate government regulations
4. Occupational Health and Workplace Safety	• Workers will be properly trained • There will be no more than 2 injuries per month • Mine should not exceed 100 degrees Fahrenheit
5. Operating Costs and Productivity	• Mine will generate over $100,000,000 per year of net profit
6. Local Community Engagement	• Community will be properly informed of mining activities and measures will be taken to minimize disturbance in accordance to the community's needs
7. Waste Management and Disposal	• Mine waste and runoff will be properly stored in an above-ground structure remaining stable for at least 200 years • Excess basalt rock will be reprocessed and sold for other industrial uses
8. Mine Closure and Rehabilitation	• Proper signage will be made around the abandoned mine • Proper measures will be made to seal off the mine, including fences surrounding the area and locked doors on all entrances

Table 1 Specifications

4.0 Conceptual Designs

In order to determine the best mining method to move forward with, we brainstormed four general mine concepts to analyze and evaluate in order to create the final mine design. When brainstorming these designs, we abided by a few general assumptions and constraints. We first decided that we would only be pursing the block caving method on the account that it was heavily recommended by our client GE and that it was overwhelmingly recommended by professionals in the field and by our independent research. We also determined that the ventilation systems in the mine would be prebuilt, and would be used by us provided it met our air contamination and safety specifications. Third, we determined that the processing plant would remain a constant as it was not the focus of this project and would only be represented by a box in our diagram with the only variations in designs being the transportation of ore to the plant.

4.1 Descriptions

4.1.1 Bucket Elevator

The first concept that we generated was for a bucket elevator. This deign worked by having load haul dumps extract the ore from the draw points and transfer them to a bucket elevator. The bucket elevators would then transfer the ore to the surface where it would be loaded onto haul trucks and transferred to the processing plant. This process gets its advantage from its generally easy and cheap process. Electric load haul dumps and an electric bucket elevator would leave generally low particulate matter in the mine itself due to the lack of diesel fuel. This design would however have more worker interaction with heavy machinery and fast moving parts, potentially leading to injuries. The use of haul trucks on the surface could also pose environmental dangers due to heavy diesel usage.

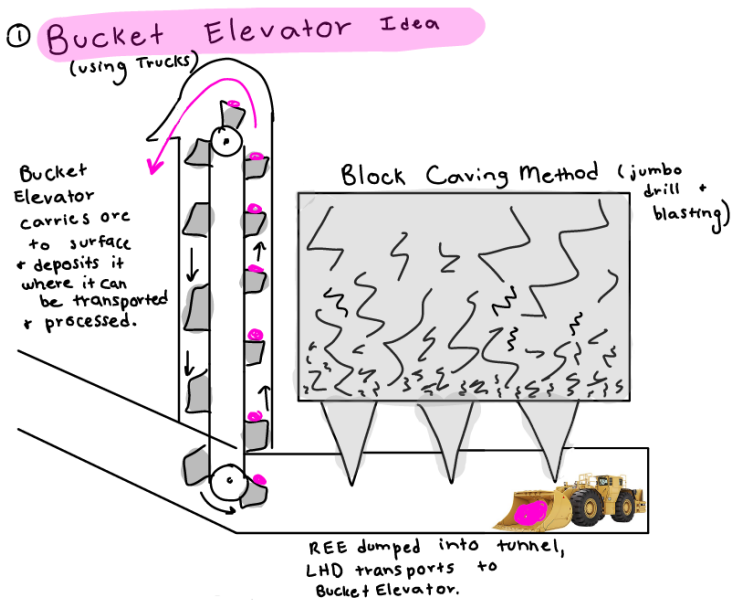

Figure 3 Bucket Elevator Sketch

4.1.2 Haul Truck

The second concept that we generated was a haul truck based design. For this design, electric load haul dumps would instead bring the ore from the draw points to various pick up locations, where it would be loaded into haul trucks, transported up a 15.3 km tunnel to the surface, and dropped off at the processing plant. This design has the advantage of being relatively simple and cheap maximizing profits and minimizing worker injury. However, because of the use of haul trucks within the mine itself, there would be a risk of diesel airborne contaminants posing a safety hazard within the mine.

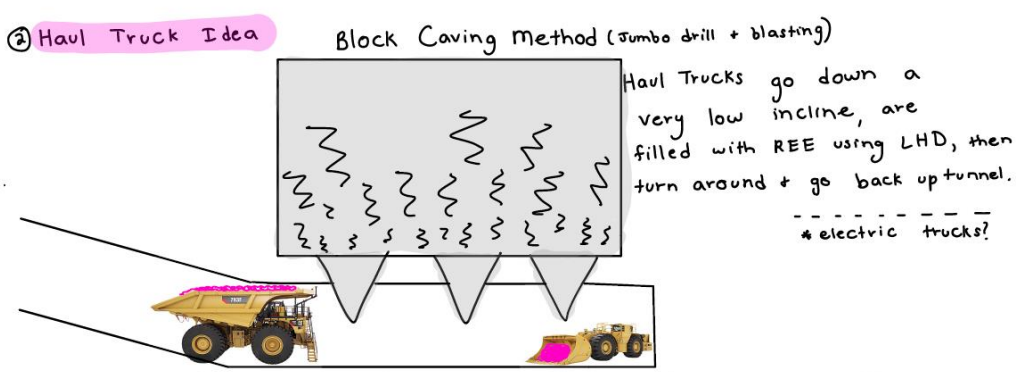

Figure 4 Haul Truck Sketch

4.1.3 Electric Train

Our third concept generated was for an electric train. Like the previous concepts, electric load haul dumps would bring the ore to a single designated drop off point on the bottom layer on the mine. From this point, the ore would be loaded into an electric train that would go up a primary straight tunnel to the surface where it would drop off the ore at the processing plant. The advantages of this design are that the train could efficiently transport large amounts of ore to the surface at once, reducing complexity and the amount of workers needed. However, we found that this design had several major disadvantages. One of these disadvantages is that the train would need a straight track and likely could not handle a grade greater than 4%, leading to a need for a very long tunnel that could be very expensive and further disturb the environment. Another major disadvantage is that the breaking of the train the majority of the way down could lead to significant heating issues within the mine, along with potential particulate pollutants.

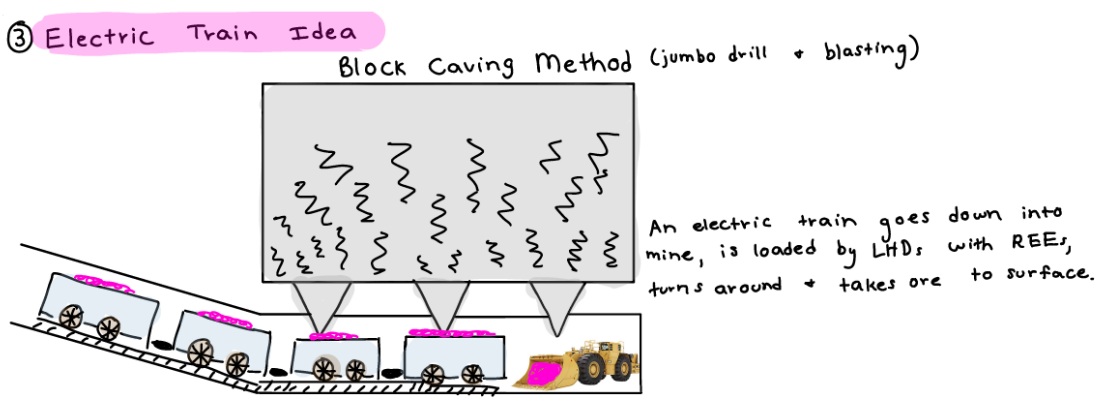

Figure 5 Electric Train Sketch

4.1.4 Conveyer Belt

Our final design was a conveyer belt that would transfer the ore to the surface. This design would work by having rock feeders push the rock onto the conveyer belt, where it would push the rock onto the conveyer belt. This conveyer belt would then conjoin with others to form a primary conveyer belt that would go up the primary tunnel to the surface of the mine to the processing plant. This concept has the advantages that it would be almost completely automated and only require workers for maintenance purposes. This concept would also eliminate almost all diesel pollution and particulates, as it would run primarily on electricity. The disadvantages of this design would be cost due to running the conveyer belts 15.3 km up the primary tunnel, along with large amounts of maintenance due to the amount of conveyer belts.

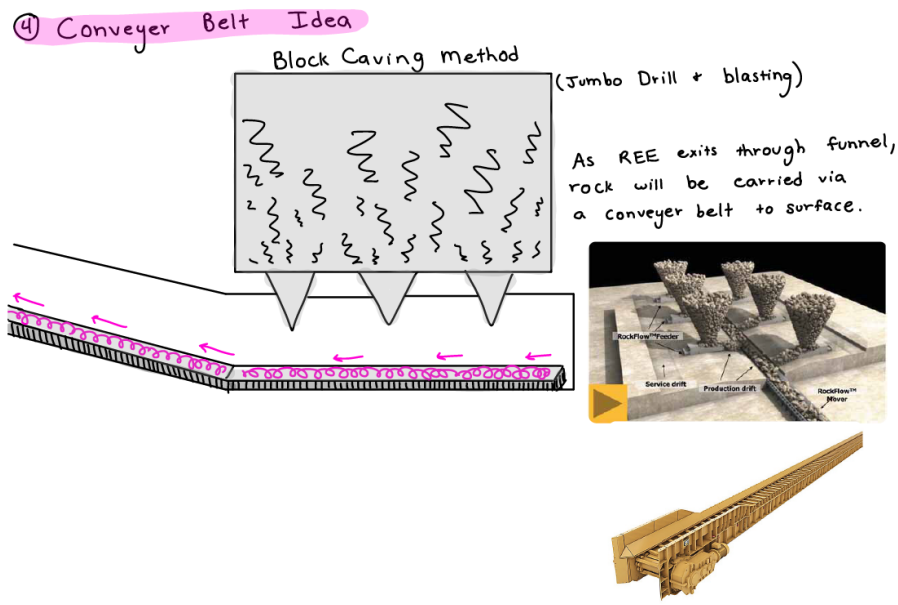

Figure 6 Conveyer Belt Sketch

4.2 Research and Analysis

In order to fully understand our design and make a well-informed decision we conducted research of the various areas of that affected our specifications. One such area of research that we conducted was in regards to airborne contamination and the exact regulations as to avoid designs that created excess amounts of these contaminates. We determined that the mine air quality must have a minimum oxygen content of 19.5% and a maximum CO2 content of 0.5%. We also determined that Carbon Monoxide (CO) cannot exceed 2.5% of the air content, Hydrogen (H) cannot exceed 0.8%, Hydrogen Sulfide (H2S) cannot exceed 0.8%, Acetylene (C2H2) cannot exceed 0.4%, Propane (C3H8) cannot exceed 0.4%, Sulfur Dioxide (S02) cannot exceed 5 ppm, and Methane (CH4) cannot exceed 1.0% (Phifer).

We also determined the general cost and capacity of some of the systems that varied between our designs. We determined that load haul dumps had a capacity of 1-25 metric tons, and that the electric versions we intended on using would cost around $1,000,000 per unit. We determined that conveyer belts could handle around 9000 tonnes of ore per hour and would cost around $4,500,000 for the entire system, and that a bucket elevator that could handle 9000 tonnes of ore per hour up 3000 meters would cost around $400,000,000. The remaining deciding factors in our concept selection were primarily estimates or specifications given by General Electric.

4.3 Concept Selection

After compiling our ideas, we generated a pairwise comparison of stakeholder requirements and weighted matrix. We decided to forgo a concept-screening table as we only had four ideas to work with, making a more comprehensive weighted matrix for all of our ideas a better solution.

	A	B	C	D	E	F	G	H	Row Total	Row Total/Total
A	1	7	6	2	2	9	5	6	38	0.28
B	1/7	1	2	1/5	1/5	4	1/2	1/4	8.29	0.06
C	1/6	1/2	1	1/5	1/7	3	1/2	1	6.51	0.05
D	1/2	5	5	1	3/2	8	4	4	29	0.21
E	1/2	5	7	2/3	1	8	3	7	32.17	0.23
F	1/9	1/4	1/3	1/8	1/8	1	1/4	1/2	2.69	0.02
G	1/5	2	2	1/4	1/3	4	1	2	11.78	0.09
H	1/6	4	1	1/4	1/7	2	1/2	1	9.06	0.07
								Total	137.5	1

1	A - Airborne Contamination
6	B - Stewardship
7	C - Environmental Management
3	D - Occupational Health and Workplace safety
2	E - Operating costs and productivity
8	F - Local Community Engagement
4	G - Waste management and disposal
5	H - Mine closure and rehabilitation

Table 2 Pairwise Comparison

	Concepts								
		A-Bucket Elevator		B-Haul Truck		C-Electric Train		D-Conveyor Belt	
Selection Criteria	Weight	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Airborne Contamination	0.28	4	1.12	3	0.84	4	1.12	4	1.12
Stewardship	0.06	3	0.18	3	0.18	3	0.18	3	0.18
Environmental Management	0.05	2	0.1	2	0.1	3	0.15	4	0.2
Occupational Health and Workplace Safety	0.21	3	0.63	2	0.42	2	0.42	4	0.84
Operating Costs and Productivity	0.23	2	0.46	2	0.46	2	0.46	2	0.46
Local Community Engagement	0.02	1	0.02	1	0.02	1	0.02	1	0.02
Waste Management and Disposal	0.09	3	0.27	2	0.18	2	0.18	2	0.18
Mine Closure and Rehabilitation	0.07	2	0.14	3	0.21	3	0.21	3	0.21
	Total Score	2.92		2.41		2.74		3.21	
	Rank	2		4		3		1	
	Continue?	yes		no		no		yes	

Table 3 Weighted Matrix

After comparing our various ideas, we determined that our two best were the bucket elevator and conveyer belt ideas. We decided that we could combine these two ideas into a design that would best adhere to the benefits of both. We decided we would use a conveyer belt system on the bottom of the mine and on the surface, and a bucket elevator to transport ore from the bottom to the surface.

5.0 Detailed Design

5.1 Defining Specifications and Profits

In order to properly define the capacities and capabilities of our mine, we compiled researched specs and those given by GE and those that we had researched, and then used those specifications to define expenses and revenue for our mine.

Many of these specifications had been defined by GE and therefore could be directly used by us in determining the mine specifications. These included the ore revenue per tonne, the cost of electricity, the ventilation costs, the workers' pay, processing costs, base costs, percentage of REE ore in the rock, and the volume of the ore deposit. As for researched specifications, we determined that a majorly automated mine such as ours would only employ around 50 workers per shift primarily for maintenance

and supervision purposes. Using specs calculated with our conveyer belts we chose, we determined a mining capacity of 9,000 tonnes per hour. The electricity usage of our mine was research through the average electricity usage of mines per tonne of ore extracted as provided by the USGS, giving us an electricity usage of 460 kWh per tonne.

The total volume of ore in our mine was calculated by multiplying the percentage of REE ore in the rock by the total volume of the rock to find a total volume of 360,943,000 cubic meters of ore. The overall weight of the ore and rock was found by determining the average density of earth's crust to be 3 tonnes per cubic meter and then multiplying this measurement by the volumes of the ore and overall rock, giving us 974,546,100 tonnes of ore and 10,258,380,000 tonnes of rock. To determine the overall operating time of our mine we multiplied the mining capacity of our mine by the total tonnes of rock that had to be extracted to find a total operating time of 1,139,820 hours, if operating at full capacity 24/7.

We next calculated our hourly revenues and expenses using these specs. While the process for calculations will be detailed, the final values can be viewed in Table 4. The total revenue for the ore was calculated simply by multiplying the revenue per tonne of ore by the ore profits per tonne provided by GE, giving us a total hourly revenue of $8,550,000.00. Hourly expenses for ventilation were determined by multiplying our hourly revenue by the ventilation cost spec, which was given as a percentage of revenue. The electricity per hour usage was determined by multiplying our electricity costs per kWh by the kilowatt usages per tonne by the mining capacity per hour. Wages per hour were simply determined by the number of workers per shift times their hourly wage. Processing costs per hour were calculated by multiplying our processing costs per tonne by the mining capacity per hour. Our base running cost per hour was determined by simply diving the overall base running cost by the number of hours our mine would be operating. Summing all of these calculations together, we found a total of $3,406,765.75 of expenses per hour, giving us an hourly net profit of $5,143,234.25 per hour.

The overall revenue of our mine was now found by multiplying our hourly revenue by the total operating hours, giving us a total of $9,745,461,000,000.00. The overall expenses were calculated simply by multiplying their respective hourly expenses by the overall operating time. In order to determine the initial expenses we found the costs of all of our major subsystems and summed them together. Summing our overall expenses, we got $3,883,509,241,068.49 of expense, giving an overall net profit of $5,861,951,758,931.51. While these numbers may seem absurd and unreasonable, they were calculated using the specs provided by GE, which assumed that all of the 9.5% of REE was uniform throughout the

entire deposit and could be completely extracted. These large profits margins also leave us large amounts of room should excessive downtime occur or if there turns out to be a major roadblock or issue in the mine development and operation.

Hourly Revenues	
Revenue from ore	$8,550,000.00

Hourly Expenses	
Ventilation	$1,710,000.00
Electricty	$496,800.00
Wages	$2,500.00
Processing	$855,000.00
Base Running Cost	$342,465.75
Total	$3,406,765.75
Hourly Net Profit	$5,143,234.25

Overall Revenue	
Overall Ore Profit	$9,745,461,000,000.00

Overall Expenses	
Ventilation	$1,949,092,200,000.00
Electricity Cost	$566,262,576,000.00
Wages	$2,849,550,000.00
Processing	$974,546,100,000.00
Base Running Cost	$390,349,315,068.49
Initial Expenses	$409,500,000.00
Total	$3,883,509,241,068.49
Net Profit	$5,861,951,758,931.51

Initial Costs			
Item	Qty	PPI	Total
Bucket Elevators	1	$400,000,000.00	$400,000,000.00
Maintmence Trucks	5	$800,000.00	$4,000,000.00
Secondary conveyer belts	5	$700,000.00	$3,500,000.00
Primary conveyer belt	1	$1,000,000.00	$1,000,000.00
Misc	1	$1,000,000.00	$1,000,000.00
Total			$409,500,000.00

General Revenue Specs	
Ore Profits ($/tonne)	$10,000.00

General Expense Specs	
Electricity Costs (Cents/kwh)	$0.12
Ventilation Costs (% of revenue	20.00%
# of workers	50
Works Pay ($/hr)	$50.00
Processing Cost ($/tonne)	$1,000.00
Base Cost ($/yr)	$3,000,000,000.00
Mine running time (hrs)	1,139,820

Mining Specs	
Mining Capacity (tonnes/hr)	9,000
Electricity Usage (kwh/tonne)	460
Electricity Consumption (kw)	4,140,000

Ore Deposit Specs	
Percentage of REE in rock	9.50%
Diameter (m)	2,200
Height (m)	1,000
Rock Volume (m^3)	3,799,400,000
Ore Volume (m^3)	360,943,000
Ore Weight (tonnes/m^3)	3
Overall Ore Weight (tonnes)	974,546,100
Rock Weight (tonnes)	10,258,380,000

Table 4 Detailed Calculations

5.2 Developing Models

Our team decided to combine an extensive network of conveyer belts with a bucket elevator to create a minimally manned and overall outstanding mining system. There are 100 total draw points in this mine, each about 60 meters apart. We chose to place them in a 10x10 grid. In between the alternating columns of draw points, as seen in figure 7, the gray-red lines represent five conveyer belts that are moving the ore pieces toward the Main Conveyer Belt. This technology is essential to the ease and promptness at which the ore can be taken to the processing plant. The conveyer belts are an

integral connection between receiving the ore from the draw points and taking it to the front of the mine.

Though it is less visible than the extensive conveyer belt system, the two-part mechanized structure called the Rock Pusher (as seen in figure 8) effectively shovels the broken rocks pieces onto the conveyer belt. The Rock Pushers are a fundamental part of the whole system because they make sure the ore is constantly flowing from the draw points to the rest of the network.

In between alternating columns of draw points where the conveyer belts are not already pushing rocks are light gray roads, as seen in figure 7. These roads are called Service Drifts, and they allow maintenance vehicles and workers access to draw points. If there are any problems with Rock Pushers being clogged or a miscellaneous rock that was not properly sent on its way to the conveyer belt system, electric-powered vehicles can access the point through the service drifts and swiftly repair the problem.

After rocks are pushed out of the draw points by Rock Pushers and are carried to the front of the mine by the extensive conveyer belt system, they are deposited onto the Main Conveyer Belt. This belt is large enough to withstand the combined mass of ore that the five other conveyer belts bring. This main belt then carries the ore perpendicular to the columns where it is picked up by the bucket elevator. As you can see in figure 9 the buckets scoop up the ore as the Main Conveyer Belt brings it, and the elevator brings it to the surface. As the bucket filled with ore turns over the top of the elevator, the ore is dumped out of a chute and onto a secondary surface-level conveyer belt, where it can then be transported to the processing plant.

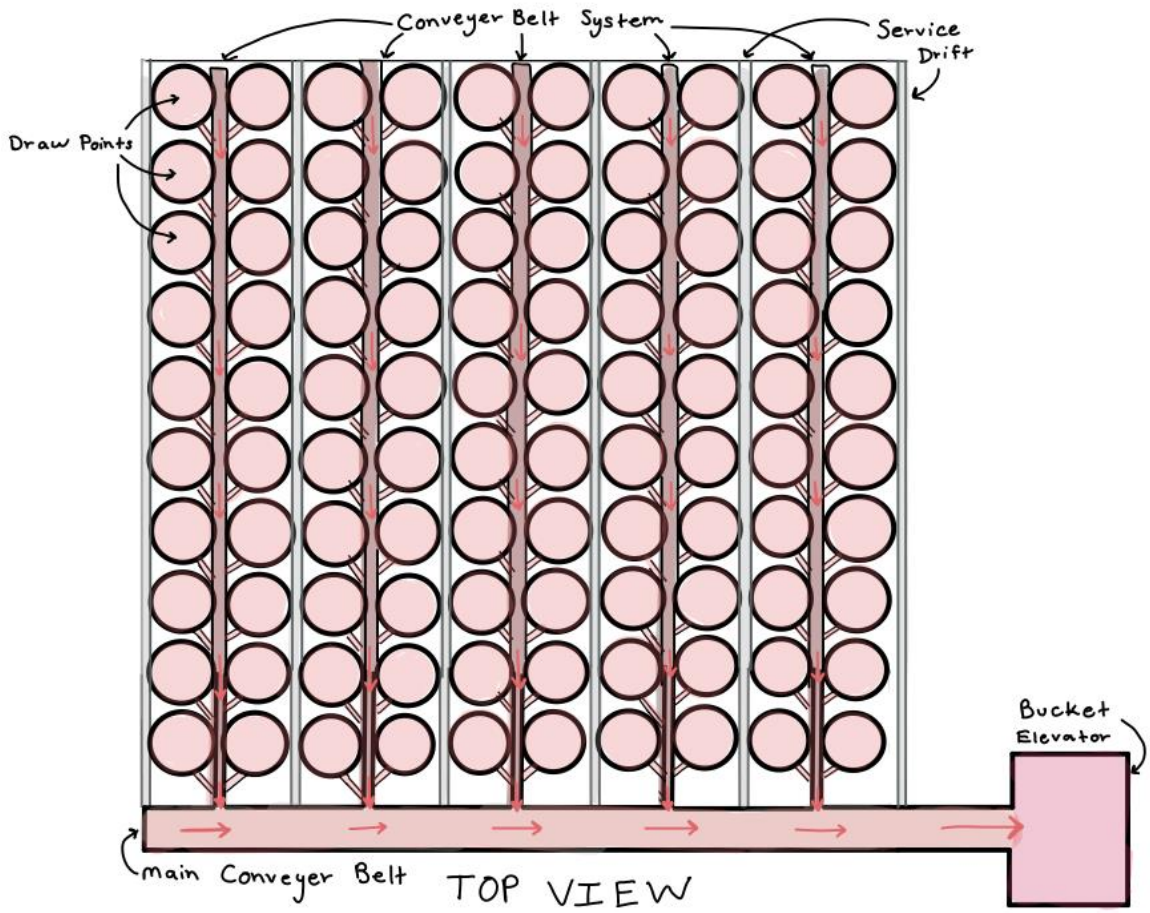

Figure 7 Top Down Diagram

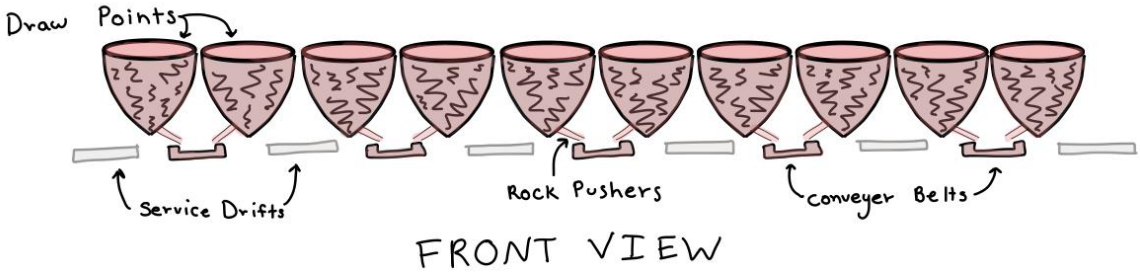

Figure 8 Front Close Up Diagram

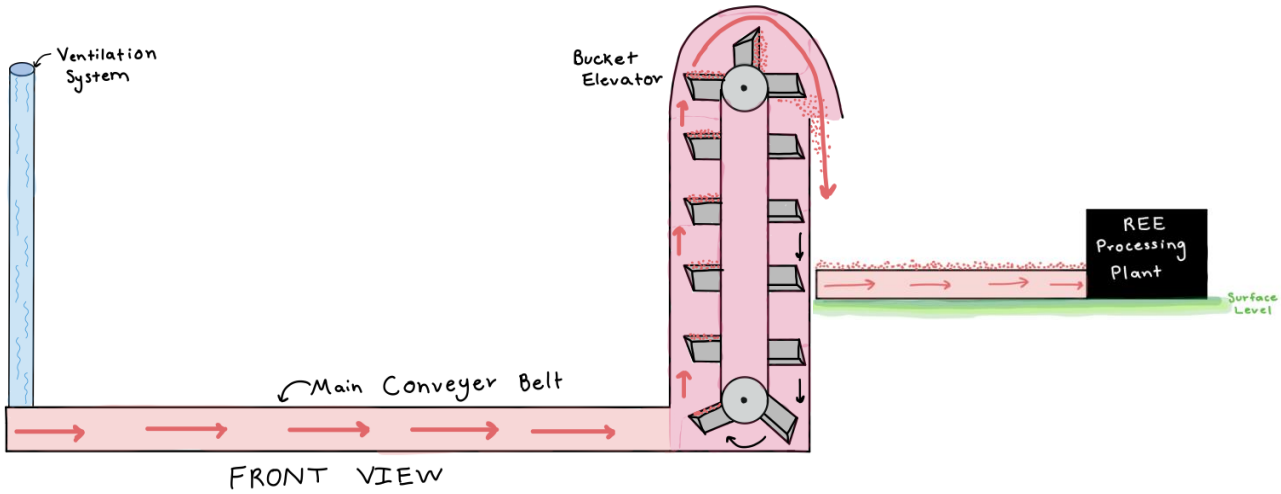

Figure 9 Front Systems View

6.0 Conclusion

Overall, GE's extraction strategy plan was a difficult task at first. However, we went through the design process systematically with the help of our Gantt chart. After determining our final design and performing calculations, we were confident that our system would be successful. We met the stakeholder needs provided by GE including meeting airborne contamination regulations and operating costs. The conveyor belt system is a success because it greatly reduces the fumes in the air that would have been increased by using trucks. Using a mostly automated mining system is important because less workers means less chances of injuring a worker. Our bucket elevator-conveyor belt system would be produced in a cost-efficient manner according to our detailed calculations, and we found our hourly net profit would be $5,143,234.25 per hour. We are happy with our design and believe it can be used as way of extracting these important Rare Earth Elements from the ore deposits in the Pocono Mountains.

7.0 References

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